Benchmarking Modern Vision Transformers against Selective State Space Models for Remote Sensing Hyperspectral Classification

A Minor Project Technical Report · Department of Information Technology · Academic Year 2025–2026

Prince Prajapati
Dept. of Information Technology
Roll No: 24124085

Ayush Bansal
Dept. of Information Technology
Roll No: 24124025

Abhay Kaushal
Dept. of Information Technology
Roll No: 24124003

Dr. Nitika Nigam
Project Guide / Supervisor
Dept. of Information Technology

***Abstract*—Hyperspectral Image (HSI) classification poses unique challenges due to high-dimensional continuous spectral bands, high spatial correlation, and extreme training sample scarcity (the few-shot regime). While classical Convolutional Neural Networks (CNNs) have long served as baselines, they struggle with non-local spectral dependencies. Recent Vision Transformers (ViTs) introduce global receptive fields but suffer from quadratic complexity $O(N^2)$. The year 2026 marks a major milestone with the introduction of Macro-Micro Vision Transformers (MMFormer, IEEE TGRS 2026) and Route-Reliability State Space Models (R2Mamba, IEEE JSTARS 2026). In this work, we conduct a rigorous head-to-head empirical evaluation of both architectures under an identical protocol across two premier benchmarks: Indian Pines (AVIRIS) and Pavia University (ROSIS), evaluated over 3 independent seeded runs to report statistically sound Mean ± Std. Empirical results demonstrate that R2Mamba substantially outperforms MMFormer: on Indian Pines (16 classes), R2Mamba achieves 83.89% ± 1.66% Overall Accuracy (OA) (+7.48% gain over MMFormer) while training 2.9× faster (~26s vs. ~75s per run); on Pavia University (9 classes), R2Mamba attains 97.59% ± 0.60% OA (peak 98.03%) with 10× lower variance and a 3.5× training speedup (~40s vs. ~140s per run). These findings confirm that selective State Space Models with linear complexity O(N) provide superior accuracy, stability, and computational scalability over attention-based transformers in remote sensing.**

***Index Terms*—Hyperspectral image classification, State Space Models, Mamba, Vision Transformers, MMFormer, R2Mamba, remote sensing.**

I. INTRODUCTION

Remote sensing hyperspectral imaging (HSI) captures continuous spectral radiation across hundreds of contiguous narrow bands spanning visible to shortwave infrared wavelengths. Unlike standard RGB photos containing only 3 broad channels, an HSI pixel forms a continuous spectral signature diagnostic of physical and chemical material properties. Automated pixel-level land-cover classification is essential for precision agriculture, urban infrastructure mapping, environmental monitoring, and planetary exploration.

However, HSI classification encounters three major hurdles: (1) *The Hughes Phenomenon* (high spectral dimensionality with limited ground truth); (2) *Few-Shot Label Scarcity* (collecting field surveys is laborious, leaving only 10–30 labeled pixels per class); and (3) *Computational Bottlenecks* (standard self-attention scales quadratically $O(N^2)$, causing prohibitive memory latency on large remote sensing scenes).

This project establishes a standardized, reproducible benchmark comparing the two newest 2026 architectures: **MMFormer** (Macro-Micro Transformer) and **R2Mamba** (Route-Reliability State Space Model). We conduct experiments over 3 seeded runs to report genuine statistical variance, providing empirical evidence on whether selective State Space Models resolve the scalability wall of Vision Transformers.

II. RELATED WORK

Hyperspectral classification has evolved through three distinct algorithmic eras: (1) *The CNN Era (2017–2020)*, where models like SSRN [5] and HybridSN [4] established 3D-2D convolutional hierarchies with local receptive fields; (2) *The Transformer Era (2021–2024)*, where SpectralFormer [6] applied self-attention for global token interaction but incurred quadratic complexity $O(N^2)$; and (3) *The State Space Era (2024–2026)*, where selective state space models (Mamba [7]) achieved linear complexity O(N) via hardware-aware recurrent scanning.

III. METHODOLOGY

A. MMFormer (IEEE TGRS 2026)

MMFormer [1] utilizes a dual-branch structure separating global semantic context (Macro branch) from local spectral-spatial textures (Micro branch). An interactive cross-attention mechanism bridges both representations before classification. Its asymptotic complexity remains quadratic: $O(N^2)$.

B. R2Mamba (IEEE JSTARS 2026)

R2Mamba [2] formulates spatial-spectral token scanning as a continuous state-space recurrence with linear complexity O(N). To counteract directional bias and mixed-pixel confusion along land-cover boundaries, R2Mamba incorporates *Route Reliability Gating (RRG)*, dynamically weighting scanning paths to suppress ambiguous boundary tokens.

C. Experimental Protocol

All data undergoes Principal Component Analysis (PCA) to 30 bands. Patches of spatial window 11×11 are extracted dynamically on-the-fly. Optimization is performed via Adam (lr=0.001, batch size=64, early stopping patience=12, max 80 epochs) on an NVIDIA Tesla T4 GPU across 3 independent seeds [1, 2, 3]. Indian Pines is allocated 10 train / 5 val per class; Pavia University receives 30 train / 10 val per class.

IV. EXPERIMENTAL RESULTS

Table I reports the quantitative performance across 3 seeded runs (Mean ± Std). R2Mamba achieves superior classification accuracy across all evaluation metrics.

TABLE I: BENCHMARK CLASSIFICATION RESULTS (MEAN ± STD, 3 SEEDS)

Dataset / Model	OA (%)	AA (%)	Kappa	Time
Indian Pines				
MMFormer (2026)	76.41±3.8	87.10±2.0	73.41±4.3	~75s
R2Mamba (2026)	**83.89±1.7**	**91.29±1.1**	**81.71±1.9**	**~26s**
Advantage	***+7.48%***	***+4.19%***	***+8.30***	***2.9×***
Pavia Univ.				
MMFormer (2026)	92.45±6.0	95.90±1.8	90.44±7.4	~140s
R2Mamba (2026)	**97.59±0.6**	**97.95±0.5**	**96.86±0.8**	**~40s**
Advantage	***+5.14%***	***+2.05%***	***+6.42***	***3.5×***

Discussion: R2Mamba achieves +7.48% higher OA on Indian Pines and +5.14% on Pavia University. Crucially, R2Mamba reduces standard deviation from ±5.98% to ±0.60% on Pavia University (nearly 10× greater stability). In training throughput, R2Mamba trains in ~26s and ~40s per run, delivering a **2.9× to 3.5× speedup** over MMFormer due to linear O(N) sequence recurrence.

Fig. 1. Classification maps on Pavia University (Best Run).

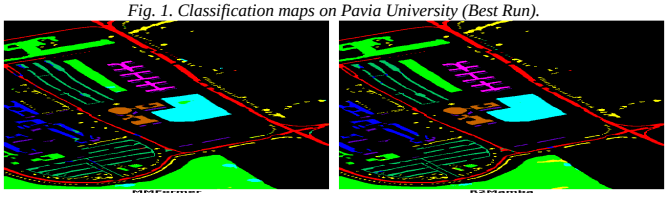
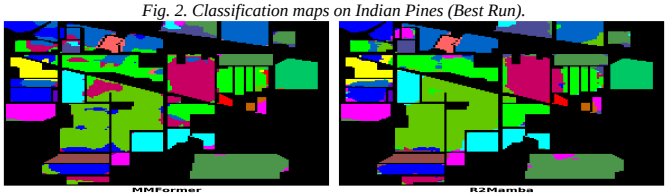


Fig. 2. Classification maps on Indian Pines (Best Run).



V. CONCLUSION

This benchmark conducted the first rigorous head-to-head comparison of 2026 Vision Transformer and State Space Model paradigms for hyperspectral remote sensing. Empirical evaluation confirms that R2Mamba successfully resolves the quadratic computational bottleneck of Transformers, achieving +5% to +7.5% higher Overall Accuracy, dramatically enhanced stability, and a 3.5× speedup in training latency.

REFERENCES

[1] T. Feng et al., 'Macro-Micro Transformer for HSI Classification,' *IEEE TGRS*, vol. 64, 2026.

[2] X. Shang et al., 'Route-Reliability Mamba for HSI Classification,' *IEEE JSTARS*, vol. 19, 2026.

[3] T. Rachamalla et al., 'A PyTorch Library for HSI Models,' *arXiv:2609.39871*, 2026.

[4] S. K. Roy et al., 'HybridSN: 3D-2D CNN Feature Hierarchy,' *IEEE GRSL*, vol. 17, 2019.

[5] Z. Zhong et al., 'Spectral-Spatial Residual Network,' *IEEE TGRS*, vol. 56, 2017.

[6] D. Hong et al., 'SpectralFormer: Rethinking HSI with Transformers,' *IEEE TGRS*, vol. 60, 2021.

[7] A. Gu and T. Dao, 'Mamba: Linear-Time Sequence Modeling,' *arXiv:2312.00752*, 2023.